RAJALAKSHMI ENGINEERING COLLEGE

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M2

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|---|---|
| **Started on** | Thursday, 17 September 2026, 10:09 AM |
| **State** | Finished |
| **Completed on** | Thursday, 17 September 2026, 10:14 AM |
| **Time taken** | 5 mins 10 secs |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 10.00 out of 10.00

**Playing with Numbers:**

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

**Example 1:**

***Input:*** *6*
***Output:****6*
***Explanation:*** *There are 6 ways to 6 represent number with 1 and 3*

*1+1+1+1+1+1*
*3+3*
*1+1+1+3*
*1+1+3+1*
*1+3+1+1*
*3+1+1+1*

**Input Format**
First Line contains the number n

**Output Format**

**Print: The number of possible ways 'n' can be represented using 1 and 3**

Sample Input

6

Sample Output

6

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

long long count_ways(int n) {

    if (n == 0) return 1;
    if (n == 1 || n == 2) return 1;

    long long w_minus_3 = 1;
    long long w_minus_2 = 1;
    long long w_minus_1 = 1;
    long long current_ways = 0;

    for (int i = 3; i <= n; i++) {
        current_ways = w_minus_1 + w_minus_3;

        w_minus_3 = w_minus_2;
        w_minus_2 = w_minus_1;
        w_minus_1 = current_ways;
    }

    return current_ways;
}

int main() {
    int n;

    if (scanf("%d", &n) == 1) {

        printf("%lld\n", count_ways(n));
    }

    return 0;
}
```

```
34
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 6 | 6 | 6 | ✔ |
| ✔ | 25 | 8641 | 8641 | ✔ |
| ✔ | 100 | 24382819596721629 | 24382819596721629 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

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